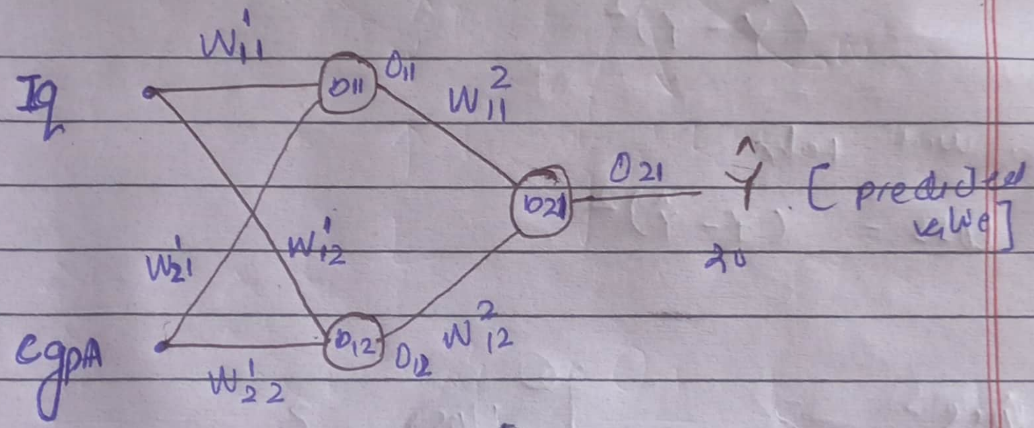
Backpropagation in Artificial Neural Networks

Day: Date:

cgpA	Iq	lpa
3	90	8
3.5	80	10
3.7	120	12
3.9	70	10



Forward Propagation

$$O_{11} = w^1_{11}\, Iq + w^1_{21}\, cgpA + b_{11}$$

$$O_{12} = w^1_{12}\, Iq + w^1_{22}\, cgpA + b_{12}$$

$$O_{21} = O_{11}\, w^2_{11} + O_{12}\, w^2_{12} + b_{21}$$

$$O_{21} = \hat{y}$$

Day: 3 Date:

The loss = (Y predicted value − Y actual value)^2

$= (Y - \hat{Y})^2$ or

$(\hat{Y} - Y)^2$

we will take

loss $= (Y - \hat{Y})^2$

(1) $\frac{d\,loss}{d w_{11}^2} = ?$

(2) $\frac{d\,loss}{d W_{11}^1} = ?$

(3) $\frac{d\,loss}{d w_{12}^2} = ?$

(4) $\frac{d\,loss}{d w_{21}^1} = ?$

(5) $\frac{d\,loss}{d b_{21}} = ?$

(6) $\frac{d\,loss}{d b_{11}} = ?$

$\frac{d\,loss}{d w_{12}^1} = ?$ $\frac{d\,loss}{d w_{22}^1} = ?$ $\frac{d\,loss}{d b_{12}}$

(1) $$\frac{\Delta Loss}{\Delta W^2_{11}} = \frac{\Delta Loss}{\Delta \hat{Y}} \times \frac{\Delta \hat{Y}}{\Delta W^2_{11}}$$

$$\frac{\Delta L}{\Delta \hat{Y}} = \frac{\Delta (Y-\hat{Y})^2}{\Delta \hat{Y}} \rightarrow -2(Y-\hat{Y})$$

$$\frac{\Delta L}{\Delta \hat{Y}} = -2(Y-\hat{Y})$$

$$\frac{\Delta \hat{Y}}{\Delta W^2_{11}} = O_{11} w^2_{11} + O_{12} w^2_{12} + b_{21}$$

$$\boxed{\frac{\Delta \hat{Y}}{\Delta W^2_{11}} = O_{11}}$$

$$\frac{\Delta Loss}{\Delta W^2_{11}} = -2(Y-\hat{Y}) \times O_{11} \rightarrow (1) \text{ find}$$

(2) $$\frac{\Delta Loss}{\Delta W^2_{12}} = \frac{\Delta Loss}{\Delta \hat{Y}} \times \frac{\Delta \hat{Y}}{\Delta W^2_{12}} \rightarrow (2)$$

$$-2(Y-\hat{Y}) \times \frac{\Delta(O_{11} w^2_{11} + O_{12} w^2_{12} + b_{21})}{\Delta w^2_{12}}$$

$$= -2(Y-\hat{Y}) \times (O_{12})$$

$$\frac{\Delta Loss}{\Delta W^2_{12}} = -2(Y-\hat{Y}) \times O_{12} \rightarrow (2)$$

(3)

$$\frac{\Delta loss}{\Delta b_{21}} = \frac{\Delta loss}{\Delta \hat{Y}} \times \frac{\Delta \hat{Y}}{\Delta b_{21}} \rightarrow (3)$$

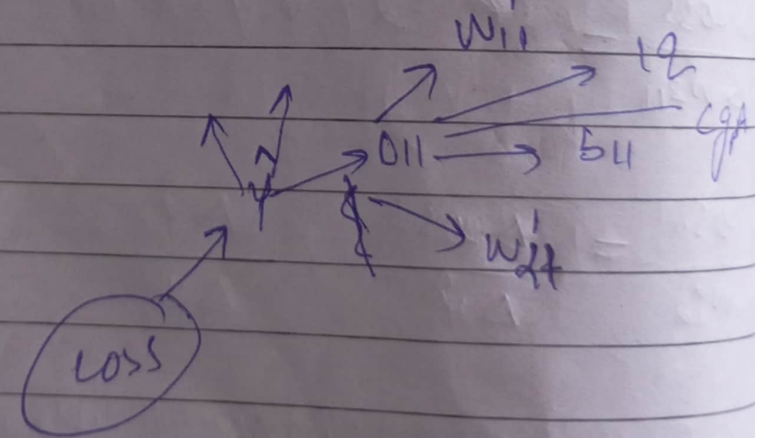
$$= -2(Y-\hat{Y}) \times \frac{\Delta(O_{11}w^2_{11} + O_{12}w^2_{21} \ldots}{\Delta b_{21}}$$

$$\frac{\Delta loss}{\Delta b_{21}} = -2(Y-\hat{Y}) \times 1$$

$$\boxed{\frac{\Delta loss}{\Delta b_{21}} = -2(Y-\hat{Y}) \times 1}$$

(4)

$$\frac{\Delta loss}{\Delta w'_{11}} = ?$$



$$\frac{\Delta loss}{\Delta w'_{11}} = \frac{\Delta loss}{\Delta \hat{Y}} \times \frac{\Delta \hat{Y}}{\Delta O_{11}} \times \frac{\Delta O_{11}}{\Delta w'_{11}}$$

$$\frac{\Delta Loss}{\Delta W^1_{11}} = -2(Y-\hat{Y}) \times \frac{\Delta(O_{11} W^2_{11} + O_{12} W^2_{12} + b_{21})}{\Delta O_{11}} \times \frac{\Delta(W^1_{11}\, Iq + W^1_{21}\, CGPA + b_{11})}{\Delta W^1_{11}}$$

$$= -2(Y-\hat{Y}) \times (W^2_{11})(Iq)$$

$$\frac{\Delta Loss}{\Delta W^1_{11}} = -2(Y-\hat{Y}) \times (W^2_{11})(Iq) \rightarrow$$

$$\frac{\Delta Loss}{\Delta W^1_{11}} = -2(Y-\hat{Y})(W^2_{11})(Iq) \rightarrow 4$$

$$\frac{\Delta Loss}{\Delta W^1_{21}} = \frac{\Delta Loss}{\Delta \hat{Y}} \times \frac{\Delta \hat{Y}}{\Delta O_{11}} \times \frac{\Delta O_{11}}{\Delta W^1_{21}} \rightarrow 5$$

$$= -2(Y-\hat{Y}) \times W^2_{11} \times (CGPA)$$

$$\frac{\Delta Loss}{\Delta W^1_{21}} = -2(Y-\hat{Y}) \times W^2_{11} \times CGPA \rightarrow 5$$

$$\frac{\Delta Loss}{\Delta b_{11}} = \frac{\Delta Loss}{\Delta \hat{Y}} \times \frac{\Delta \hat{Y}}{\Delta O_{11}} \times \frac{\Delta O_{11}}{\Delta b_{11}}$$

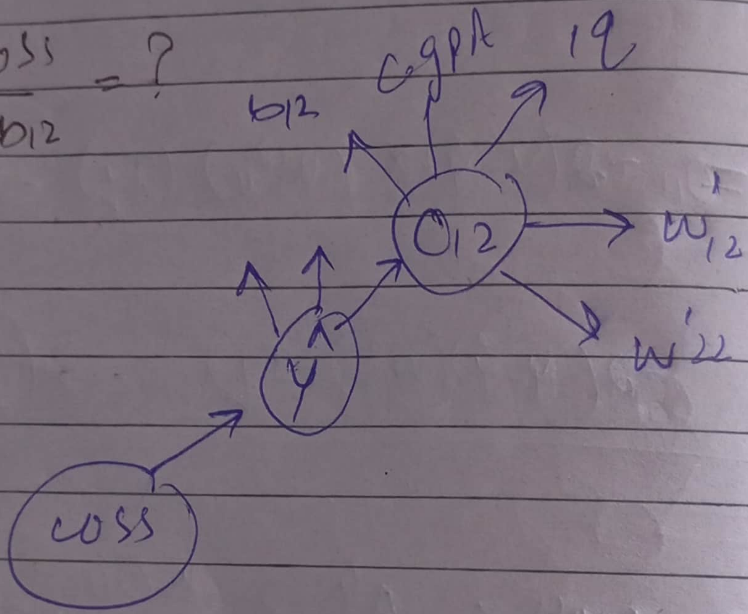
$$\frac{\Delta Loss}{\Delta b_{11}} = -2(Y-\hat{Y}) \times (W^2_{11}) \times (1)$$

$$\frac{\Delta Loss}{\Delta b_{11}} = \boxed{-2(Y-\hat{Y})(W^2_{11})(1)}$$

(7) $\frac{\Delta loss}{\Delta w'_{12}} = ?$

(8) $\frac{\Delta loss}{\Delta w'_{22}} = ?$

(9) $\frac{\Delta loss}{\Delta b_{12}} = ?$



(7) $$\frac{\Delta loss}{\Delta w'_{12}} = \frac{\Delta loss}{\Delta \hat{y}} \times \frac{\Delta \hat{y}}{\Delta O_{12}} \times \frac{\Delta O_{12}}{\Delta w'_{12}}$$

$$= -(y-\hat{y}) \times \frac{\Delta (O_{11} w^2_{11} + O_{12} w^2_{11} + b_2)}{\Delta O_{12}}$$

$$\times \frac{\Delta (w'_{12}\, iq + w'_{22}\, cgpA + b_{12})}{\Delta w'_{12}}$$

$$\frac{\Delta loss}{\Delta w'_{12}} = -(Y-\hat{Y})(w^2_{11})(iq) \rightarrow 7$$

$$\frac{\Delta loss}{\Delta w'_{22}} = -(Y-\hat{Y})\frac{\Delta \hat{Y}}{\Delta O_{12}} \times \frac{\Delta O^{12}}{\Delta w'_{22}}$$

$$= -(Y-\hat{Y}) \times w^2_{11} \times cgpA)$$

$$= -(Y-\hat{Y}) \times w^2_{11} \times cgpA)$$

$$= -(Y-\hat{Y})\, w^2_{11}\, cgpA$$

$\rightarrow 8$

$$\frac{\Delta loss}{\Delta b_{12}} = ?$$

$$= -(Y-\hat{Y})\frac{\Delta \hat{Y}}{\Delta O_{12}} \times \frac{\Delta O^{12}}{\Delta b_{12}}$$

$$= -(Y-\hat{Y})(w^2_{11})(1)$$

$$= -(Y-\hat{Y})(w^2_{11})(1)$$

$$\frac{\Delta loss}{\Delta b_{12}} = -(Y-\hat{Y})(w^2_{11})(1) \rightarrow 9$$

Day: Date:

1

$$\frac{d\,loss}{d\,w^2_{11}} = -2(y-\hat{y})\,O_{11}$$

2

$$\frac{d\,loss}{d\,w^1_{11}} = -2(y-\hat{y})(w^2_{11})(IQ)$$

3

$$\frac{d\,loss}{d\,w^2_{12}} = ?\ -2(y-\hat{y})\,O_{12}$$

4

$$\frac{d\,loss}{d\,w^1_{21}} = -2(y-\hat{y})(w^2_{11})(CGPA)$$

5

$$\frac{d\,loss}{d\,b_{21}} =$$

Day:

6

$$\frac{d\,loss}{d b_{11}} = -2(Y-\hat{Y})(w^2_{11})(1)$$

7

$$\frac{d\,loss}{d w^1_{12}} = -(Y-\hat{Y})(w^2_{11})(IQ)$$

8

$$\frac{d\,loss}{d w^1_{22}} = -(Y-\hat{Y})\, w^2_{11}\, CGPA$$

9

$$\frac{d\,loss}{d b_{12}} = -(Y-\hat{Y})(w^2_{11})(1)$$

The End